

```
In [12]: from sklearn.neighbors import KNeighborsClassifier

In [13]: knn = KNeighborsClassifier(n_neighbors = 1)

In [14]: print(knn)
KNeighborsClassifier(algorithm='auto', leaf_size=30, metric='minkowski',
                    metric_params=None, n_jobs=1, n_neighbors=1, p=2,
                    weights='uniform')

In [15]: knn.fit(X,y)

Out[15]: KNeighborsClassifier(algorithm='auto', leaf_size=30, metric='minkowski',
                    metric_params=None, n_jobs=1, n_neighbors=1, p=2,
                    weights='uniform')

In [16]: knn.predict([3,5,4,2])

C:\Users\Sony\Anaconda3\lib\site-packages\sklearn\utils\validation.py:395: DeprecationWarning: Passing 1d arrays as data is deprecated in 0.17 and will raise ValueError in 0.19. Reshape your data either using X.reshape(-1, 1) if your data has a single feature or X.reshape(1, -1) if it contains a single sample.
  DeprecationWarning)
```

Figure 3: Code for Jupyter notebook 3

```
Out[16]: array([2])

In [17]: X_new = [[3,5,4,2],[5,4,3,2]]
          knn.predict(X_new)

Out[17]: array([2, 1])

In [18]: from sklearn.linear_model import LogisticRegression
          logreg = LogisticRegression()
          # fit the model
          logreg.fit(X,y)
          #predict the response
          logreg.predict(X_new)

Out[18]: array([2, 0])
```

Figure 4: Code for Jupyter notebook 4

```
In [19]: print(iris.data[2,0])
4.7

In [20]: # store the predicted response values
          y_pred = logreg.predict(X)

          # check how many predictions were generated
          len(y_pred)
          from sklearn import metrics
          print(metrics.accuracy_score(y,y_pred))

0.96
```

Figure 5: Code for Jupyter notebook 5

Once this is done, check for the accuracy of the model. In this case, the model has provided an accuracy of 0.96, which is 96 per cent, and is considered good. Please see Figure 5.

As stated earlier, Scikit-learn, along with the combination of SciPy and NumPy, is one of the best options available today for machine learning predictions. **END** 

References

- [1] <https://stats.stackexchange.com/questions/70761/what-is-the-difference-between-online-and-batch-learning>
- [2] https://en.wikipedia.org/wiki/Reinforcement_learning
- [3] https://en.wikipedia.org/wiki/Online_machine_learning
- [4] <http://scikit-learn.org/stable/tutorial/basic/tutorial.html>

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